

Lesson Notes

MODULE 1 | LESSON 1

SYSTEMIC RISK: MACROECONOMIC INDICATORS

Reading Time	30 minutes
Prior Knowledge	The Great Financial Crisis
Keywords	Non-stationarity, Shadow banking, Disintermediation, Systemic risk, Macroprudential measures, Microprudential measures, Emergent property, Complexity, Network, Contagion

In Portfolio Theory, we discussed portfolio construction as a balancing act between profit maximization and drawdown or loss minimization. The portfolio theories consisted of relatively straightforward models of return dynamics and were mostly focused on equities. Moreover, the approaches were largely premised on the persistence of basic historical statistical phenomena. In this course, we expand our definition of risk several times over. We accomplish this not only by taking into account additional asset classes such as credit and traded volatility products but we also step back to survey the financial ecosystem as a whole (that is, as a network) and appreciate the types of risk that may threaten this delicate system. More to the point, we take up the various methods for quantifying these risks and managing them.

1. Financial Engineering as Risk Management

Engineering might be considered the science of problem solving. More specifically, it is the science of problem solving with technical and quantitative tools, which include various models and algorithms. In fact, as we have seen over the course of the program, we find ourselves solving disparate financial problems by applying a growing set of tools, which themselves are evolving. The problem of risk is in many ways the perfect challenge for the financial engineer. Does not the following description of systemic risk management sound like an accurate description of financial engineering's more general endgame?

"A practical implementation" that translates "economic concepts into very particular choices: one must decide which attributes of which entities will be measured, how frequently and over what observation interval, and with what levels of granularity and accuracy. Summary measures involve further choices on how to filter, transform, and aggregate the raw inputs" (Bisias 2).

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domain to practice the type of technical thinking that we have been learning throughout this program. This is not to say that risk management is not, in so many ways, a uniquely vital and formidable challenge. It is only to say that the engineering of risk management is not essentially different from the engineering we have been learning to date. Conversely, the critical thinking and quantitative tools applied to the assessment of a wide range risks should be kept in mind for potential application in other areas of finance. Throughout the rest of this course, try to stay alert to the accomplishment of both of these objectives.

2. The Curse of Non-Stationarity

Risk managers are called to play several roles. Certainly, they must make “backward-looking” predictions, such as Value at Risk based on historical data, whether this is a non-parametric simulation based on past data points, a parametric simulation based on a past distribution, or an expected distribution based on past observations. There are many quantitative tools to employ for these tasks, and we have examined many of these in the last and previous courses—and we will again in this course as well.

We can learn a lot from history, and from historical data specifically. However, assuming with too much confidence that the future will resemble the past can be dangerous. For this reason, the historically focused, backward-looking type of risk management is often disparaged as “driving by looking in the rear-view mirror.” So a good risk manager, like a good portfolio manager, must also be looking _forward: _at the emerging possibilities and probabilities that do not have a direct precedent in the historical data. We focus much more on this type of risk management in this course. The subject of the current lesson, the measurement and management of systemic risk, belongs mostly to this forward-looking category of responsibilities. Underlining this point, Bisias et al. go so far as to say, “In fact, the very notion of systemic risk is a good illustration of non-stationarity” (29).

3. Diversity, Complexity and Networks

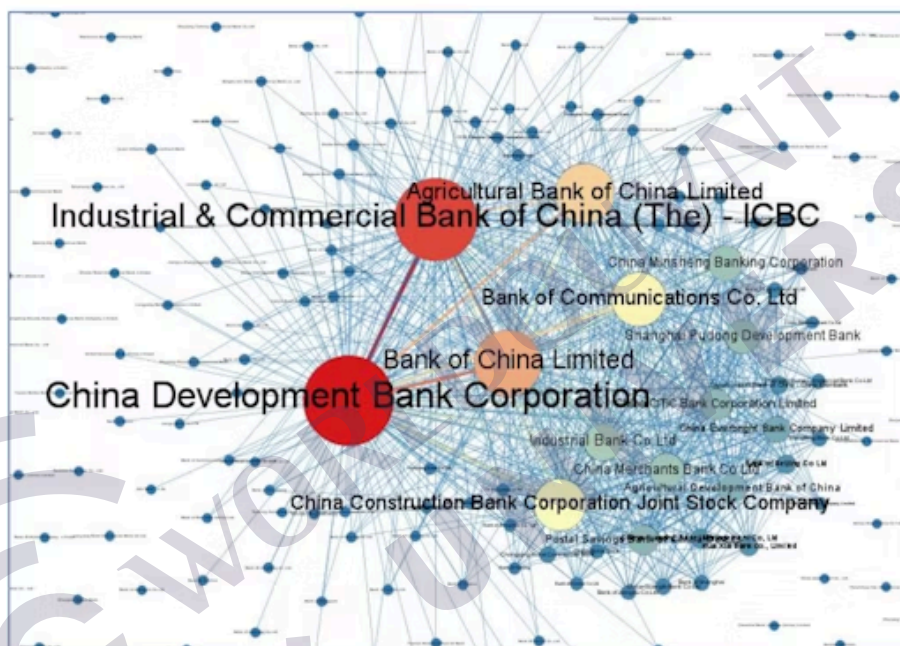
As Bisias et al. point out, systemic risk may take any of several forms, and it has many definitions, and due to the infrequency of systemic shocks, we have a precious little data to support our investigations (2, 4). As we witnessed in the application of evolutionary algorithms in previous courses, these competing models and definitions of systemic risk can actually enrich our overall understanding. Indeed, Bisias et al. announce the need for “a robust framework for monitoring and managing financial stability,” which “must incorporate both a diversity of perspectives and a continuous process for re-evaluating the evolving structure of the financial system and adapting systemic risk measures to these changes” (2). In this sense, diversity of thought and evolutionary progress go hand in hand.

One aspect of the challenge of defining, recognizing, or measuring systemic risk is that it is often the result of the accumulation of particularities that by themselves would not pose a risk. On this point, Bisias says that “the fallacy of composition applies: patterns exist in market dynamics at the system level that are distinct from the simple aggregation of the behavior of the individual

about how "legal entities and the financial contracts between them are basic building blocks for understanding and modeling systemic risk" (Bisias '57) and how Network Graphs help us visualize such a model. Note that the upper and lower network graphs resemble each other, but look more closely to observe how many more connections appear in the lower network graph, indicating that the Chinese interbank system is becoming increasingly dense with each passing year. A network graph of interbank connections at the global or some other regional level would not look much different and would certainly exhibit the same worrying trend toward densification. Such visualizations are especially helpful when discussing systemic risk because they highlight the potential for contagion in the financial system: Without having direct counterparty exposure to any particular market participant, most of the banks shown still have a level of indirect exposure to most of the other banks. In the aggregate, this secondary or tertiary indirect exposure could be massive. You should expect to see in this course much more discussion of graph and network theory as tools for understanding the transmission of risk. We will use these tools not just for visualizing the possibilities but for estimating the probabilities of events based on conditioning events which are modeled as nodes on a similar graph.

Figure 1: Chinese Interbank Connections Network





Note(s): Upper and lower graphs illustrate the interbank connections in the Chinese interbank market in 2014 and 2017 respectively

Source: Qi, Ming, et al. "Interconnectedness and Systemic Risk Measures of Chinese Financial Institutions." *Kybernetes*, 2022, vol. 51. no. 13, pp. 57–81, <https://www.emerald.com/insight/content/doi/10.1108/K-04-2021-0270/full/html>.

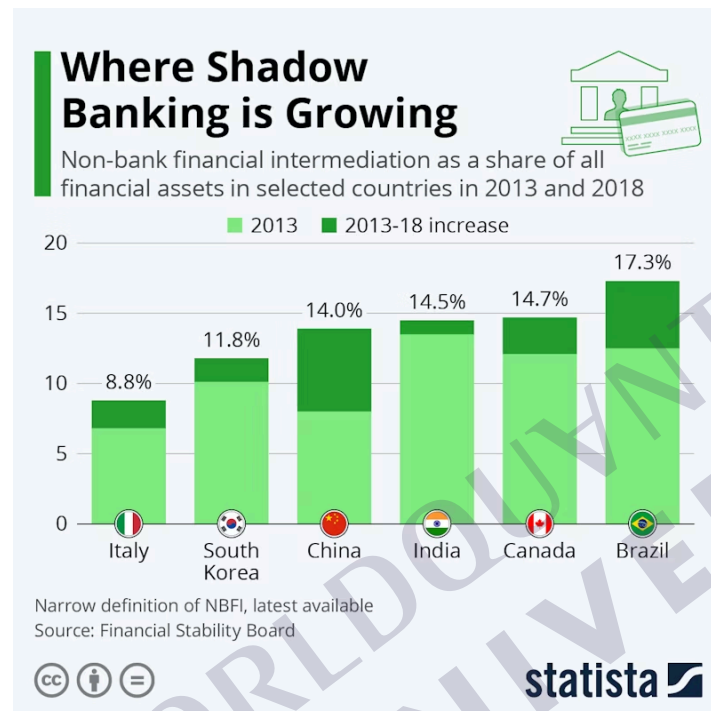
4. Shadow Banking

Note too in the above graphic that only inter_bank_ connections are shown; we do not see the asset managers, hedge funds, or other investment firms. There would be much too many such entities to fit in a single graphic. Adding the connections between all of these market participants would mean showing such a dense layer that there would hardly be any white space left in the middle of the graphic.

These other financial entities more or less form what is referred to as the **shadow banking** sector in the reading. It must be said that this is a somewhat loaded description: Activity that occurs "in the shadows" usually implies that the activity is unethical, illegal or otherwise inappropriate. There are many definitions of shadow banking (Schwarz 620–1), some of which may in fact include this type of characterization. However in its strictest sense, it simply refers to when non-bank entities

regulatory scrutiny as banking in most jurisdictions. So it follows, according to Bisias et al., that "by moving assets off the balance sheets of highly regulated, traditional depositories, and into less regulated" entities, there is the potential for risks, including systemic risk, to emerge unseen by regulators or other market participants.

Figure 2: Growth of Disintermediation Around the World



Source: Buchholz, Katharina. "Where Shadow Banking is Growing." *Statista*, 18 Nov 2020, <https://www.statista.com/chart/23534/nbfi-growth-by-country/>.

As these less-regulated and unregulated entities provide more and more financing (as shown in figure 2), they become increasingly important to the financial system as well as a growing source of potential contagions, all the more so since they operate outside the purview of regulators.

5. Conclusion

In this module, we started our journey to comprehend and measure the many risks that lurk in the financial ecosystem. We went beyond the standard market risk measures that focus on equity prices and Value at Risk. In the next module, we continue this journey with the second half of the same reading, where we gain exposure to network analysis, stress testing and scenario analysis, to name just a few of the exciting approaches to come.

References

- Bisias, Dimitrios, et al. "A Survey of Systemic Risk Analytics." *Annual Review of Financial Economics*, vol. 4, no. 1, Oct 2012, pp. 255–296, https://dspace.mit.edu/bitstream/handle/1721.1/87772/Lo_A%20survey.pdf?sequence=1&isAllowed=y.

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